

EEE 554 (79766, 84963)
Probability and Random Processes
Fall 2026

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Office Hours: **2:30–4:00 pm Tuesdays and by appointment**
Zoom: <https://asu.zoom.us/j/2520846968> (passcode: chaos)
In person: GWC 610

Recitation sessions via Zoom: Tuesdays 4:30-5:30 pm (on dates posted on Schedule)
Zoom link: <https://asu.zoom.us/j/2520846968> (passcode: chaos)

Please note: We will not meet in SCOB 252. The recitation classes will be online throughout the semester.

Prerequisites

1. EEE 203 Signals and Systems or an equivalent lower-division undergraduate course;
2. EEE 350 Random Signal Analysis at ASU or an equivalent upper-division undergraduate course that covers probability and the fundamentals of one and multiple random variables;
3. An undergraduate course in *Linear Algebra*;
4. Thorough knowledge of one-variable and multi-variable calculus.

Required Textbook

Steven Kay, *Intuitive Probability and Random Processes using MATLAB*, Springer, 2006 (electronic copy available at ASU Library Resource Organizer).

Required Software

Matlab (access at MyASU apps)

Course aim

The course is designed to study methods for processing signals that have uncertain values. Although mathematical in nature, the study is applicable in many real-world applications involving signals from noisy sensor measurements or key parameters extracted from random observations.

Course learning objectives

At the completion of this course, you will be able to

1. demonstrate a thorough knowledge of basic probability theory and describe the key elements contributing to the random nature of an event,
2. use joint probability and marginal densities of multiple random variables to calculate mean, covariance and correlation,
3. identify properties of transformed random variables,
4. process sampled sensor measurements as random vectors and apply limit theorems to simplify analysis,
5. differentiate between stationary, wide sense stationary (WSS) and non-stationary random processes,
6. evaluate the temporal or ergodic mean estimates and power spectral density of a WSS random process,
7. analyze the output signal of a linear time-invariant system whose input is a WSS random process embedded in additive white Gaussian noise,
8. use the Wiener filter to minimize the mean-squared estimation error of an observed random process,
9. use Markov chains and transition matrices to model random processes whose current state depends only on the immediate previous state, and
10. apply theory of stochastic random processes to real-world problems.

Course outline

Chapters from the required textbook

- *Chapters 1-4:* Review of probability theory, axioms, experiments, outcomes, conditional probability, independence, Monte Carlo simulations
- *Chapters 5, 6, 10, 11:* Random variable, continuous and discrete classes, cumulative distribution function, probability density and mass functions, function of a random variable, moments, characteristic functions, estimation of mean and variance
- *Chapters 7, 8, 12, 13:* Multiple random variables, joint distribution and density functions, marginal statistics, functions of two random variables, joint moments and characteristic functions, conditional distributions
- *Chapters 9, 14, 15:* Random vectors, law of large numbers, central limit theorem

- *Chapters 16-19:* Random processes, stationary and nonstationary random processes, wide sense stationary random processes, power spectral density, linear systems with stochastic inputs, multiple random processes
- *Chapters 20, 21, 22:* Gaussian random process, Wiener random process, Poisson random process, Markov processes and Markov chains

Grading

- Homework assignments (six times): 15%
- Exam 1: 20% (Modules 1 and 2)
- Exam 2: 20% (Modules 3 and 4)
- Special Reading Assignment: 5% with the opportunity to earn 5% extra credit
- Final exam: 40% (Modules 1-6)
- Total possible points: 105
- Method for determining Semester Letter Grade: Grading on a curve (e.g., A⁺ for the highest grade out of 105 possible points)

Homework and exam information

- All homework assignments must be submitted electronically on Canvas before the deadline. **Late homework will not be accepted.**
- Exams are closed book and closed notes. Questions are designed to test your understanding of the material; they are similar to assignments and examples but different enough to test comprehension and problem solving skills.
- Items allowed during exams: (a) scientific calculator; (b) writing supplies (e.g., pen, pencil, eraser, ruler); (c) a letter-size paper formula sheet prepared by each individual student (one side only for each midterm exam and two sides for the final exam). The formula sheet will be collected with each exam; they may include important equations but preferably not worked out problems or book examples.
- Items not allowed during exams: books, lecture material, notes, handheld wireless devices (e.g., tablet, smart phone, laptop, wearable computer such as smartwatch, personal digital assistant).
- No cheating on exams will be tolerated.
- Both online and hybrid students will use the Honorblock remote proctoring system for exams. Information on Honorblock will be posted on Canvas.

Generative AI Generative AI is a technology that can often be useful in helping students learn the theories and concepts in this course. However, unless explicitly allowed by your instructor, the use of generative AI tools to complete any portion of a course assignment or exam will be considered academic dishonesty and a violation of the ASU Academic Integrity Policy. Students confirmed to be engaging in non-allowable use of generative AI will be sanctioned according to the academic integrity policy and FSE sanctioning guidelines.

Academic integrity All engineering students are expected to adhere to the ASU Student Honor Code and the ASU academic integrity policy, which can be found at <https://provost.asu.edu/academic-integrity/policy>).

Students are responsible for reviewing this policy and understanding each of the areas in which academic dishonesty can occur. If you have taken this course before, you may not reuse or submit any part of your previous assignments without the express written permission from the instructor.

All student academic integrity violations are reported to the Fulton Schools of Engineering Academic Integrity Office (AIO). Withdrawing from this course will not absolve you of responsibility for an academic integrity violation and any sanctions that are applied. The AIO maintains a record of all violations and has access to academic integrity violations committed in all other ASU college/schools.

Student copyright responsibilities Students must refrain from uploading to this course shell, discussion board, website used by the course instructor or any other course forum, material that is not your own original work, unless you first comply with all applicable copyright laws. Course instructors reserve the right to delete materials from the course shell on the grounds of suspected copyright infringement.

The contents of this course, including lectures and other instructional materials, are copyrighted materials. Students may not share outside the class, including uploading, selling or distributing course content or notes taken during the conduct of the course. Any recording of class sessions by students is prohibited, except as part of an accommodation approved by the Disability Resource Center. (see ACD 304–06, "Commercial Note Taking Services" and ABOR Policy 5-308 F.14 for more information).

Policy against threatening behavior, per the Student Services Manual, SSM 104–02 Students, faculty, staff, and other individuals do not have an unqualified right of access to university grounds, property, or services (see SSM 104-02). Interfering with the peaceful conduct of university-related business or activities or remaining on campus grounds after a request to leave may be considered a crime. All incidents and allegations of violent or threatening conduct by an ASU student (whether on- or off-campus) must be reported to the ASU Police Department (ASU PD) and the Office of the Dean of Students.

Disability accommodations Suitable accommodations are made for students having disabilities. Students needing accommodation must register with the ASU Student Accessibility and Inclusive Learning Services office and provide documentation of that registration to the instructor. Students should communicate the need for an accommodation in enough time for

it to be properly arranged. See ACD 304-08 Classroom and Testing Accommodations for Students with Disabilities.

Harassment and sexual discrimination Arizona State University is committed to providing an environment free of discrimination, harassment, or retaliation for the entire university community, including all students, faculty members, staff employees, and guests. ASU expressly prohibits discrimination, harassment, and retaliation by employees, students, contractors, or agents of the university based on any protected status: race, color, religion, sex, national origin, age, disability, veteran status, sexual orientation, gender identity, and genetic information.

Title IX is a federal law that provides that no person be excluded on the basis of sex from participation in, be denied benefits of, or be subjected to discrimination under any education program or activity. Both Title IX and university policy make clear that sexual violence and harassment based on sex is prohibited. An individual who believes they have been subjected to sexual violence or harassed on the basis of sex can seek support, including counseling and academic support, from the university. If you or someone you know has been harassed on the basis of sex or sexually assaulted, you can find information and resources at <https://sexualviolenceprevention.asu.edu/faqs>.

Photo requirement Arizona State University requires each enrolled student and university employee to have on file with ASU a current photo that meets ASU's requirements. ASU uses your Photo to identify you, as necessary, to provide you educational and related services as an enrolled student at ASU. If you do not have an acceptable Photo on file with ASU, or if you do not consent to the use of your photo, access to ASU resources, including access to course material or grades (online or in person) may be negatively affected, withheld or denied.

Syllabus disclaimer The syllabus is a statement of intent and serves as an implicit agreement between the instructor and the students. If unforeseen events mandate syllabus changes, the syllabus will be updated and posted on Canvas.